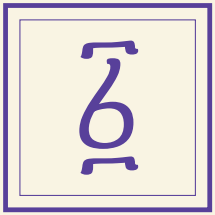
# From a language model to a planning agent

*Weekend 1: Large language models*

Carlos Cotrini
*AI in Industry, ETH Zurich*
*Weekend 1, October 2026*

# From a language model to a planning agent





*Section 1*

# Supervised fine-tuning

# A pretrained model continues your text

| *You write* | *It writes next* |
| --- | --- |
| Dear customer, thank you for | your order. We are pleased to confirm that it ships on Monday. |
| The central bank raised interest rates on | Thursday, citing inflation that stays above its target. |
| What is the capital of Switzerland? | What is the capital of Germany? What is the capital of Italy? |

*It continues the kind of page it thinks it is on.*

*Invented examples*

# People write the answers we want

*Prompt* *An invented example*

Explain what a mortgage is to a teenager.

*Reply, written by a person*

A mortgage is a loan for buying a home. You pay it back in small amounts every month.

*And many more examples like it*



*People write the replies*

*InstructGPT, 2022: 12 725 prompts, replies by about 40 contractors. LIMA, 2023: 1 000 examples.*

# Game 3's learning, on new text

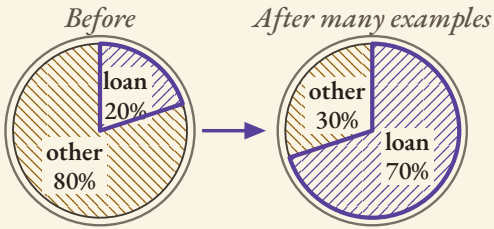
*What the model has so far*

Explain what a mortgage is to a teenager. **A mortgage is a**

*The next word of the reply*

loan

*Game 3's training: what came before, and the word that came next.*




*Invented chances, for illustration*

# A conversation is one long prefix

*What you see*

| | |
|---|---|
| *System* | You are a helpful assistant of our bank. |
| *You* | Explain what a mortgage is to a teenager. |
| *Assistant* | A mortgage is a loan for buying a home. |

*What the model reads*

`[system]` You are a helpful assistant of our bank.
`[user]` Explain what a mortgage is to a teenager.
`[assistant]` **A mortgage is a loan for buying a home.** `[end]`



*The markers differ from one model to another.*



*Every new turn sends the whole conversation again.*

# Fine-tuning teaches a manner of answering

*You ask*

What is our refund policy?

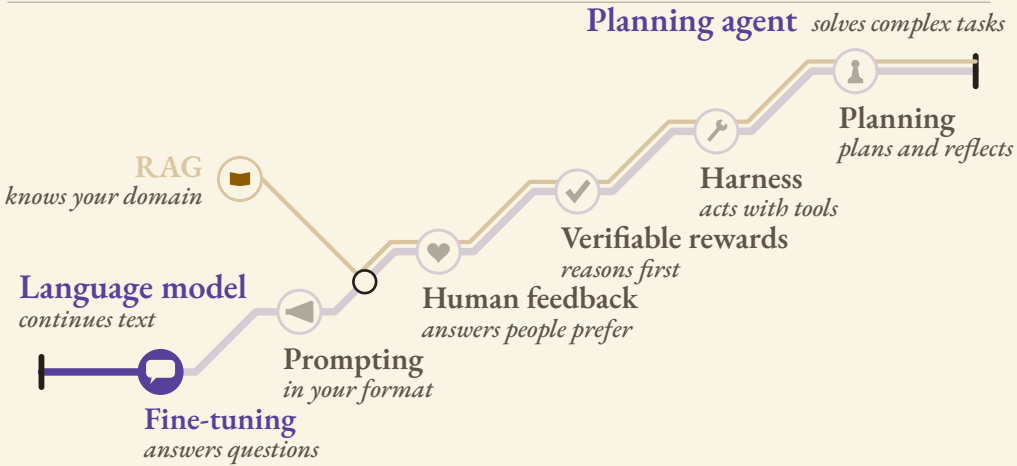
*The chat model answers*

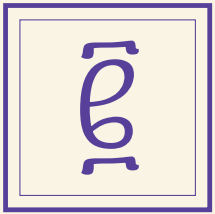
Thank you for asking! Our policy is simple: you can return any item within a month for a full refund.

*An invented answer*

- ***The manner.*** *Polite, short, in the format you expect.*
- ***The facts.*** *The model never saw our policy. It writes the words that sound right.*
- *Facts come from pretraining, or from documents you give it.*

# One rung up: a chat model





*Section 2*

# Prompt engineering

# Different words in, a different answer out

*You ask*

Explain what a mortgage is.

Explain what a mortgage is to a teenager, in two sentences.

Explain what a mortgage is, as a table with two columns.

*The same model answers*



A mortgage is a loan for a home. You pay it back monthly.

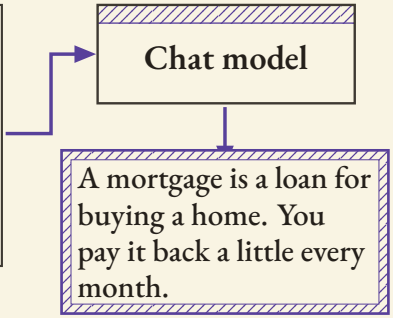
| Term | Meaning |
|---|---|
| Loan | Money you borrow |
| Interest | The price of the loan |

*Invented answers*

*The model is the same in all three rows. Only the words that go in are different.*

# A prompt is a brief to a new colleague

| | |
|---|---|
| *Role* | You are a bank advisor. |
| *Task* | Explain what a mortgage is. |
| *Example* | Like this: a loan is money you borrow and pay back. |
| *Format* | Two sentences, in plain words. |




*A capable new colleague has read everything and knows nothing about your company.*

*An invented answer*

# Examples in the prompt teach by showing

*The prompt*

*Message* The parcel arrived broken. *Label* Complaint

*Message* Thank you, great service! *Label* Praise

*Message* Where is my order? *Label*

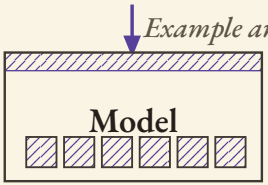
**Question**
*The model draws*



*Nothing inside the model has changed. The examples have to be in the prompt every time.*

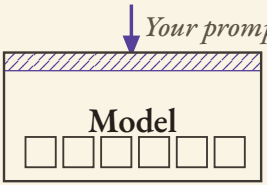
# A prompt steers the model without changing it

## Fine-tuning




The numbers inside change, and so do all later answers.

## Prompting

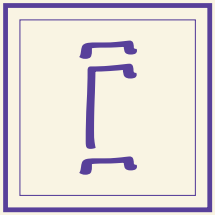



The numbers inside stay the same. Only the input differs.

*Prompting is quick to try and training is slow. Start with the prompt, and train when it is not enough.*

# Two rungs up: your words steer the model

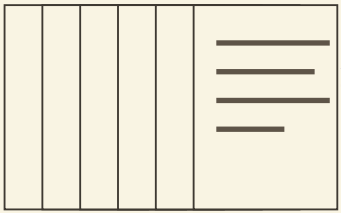




*Section 3*

# RAG

# A prompt holds only part of the archive



*All our documents*

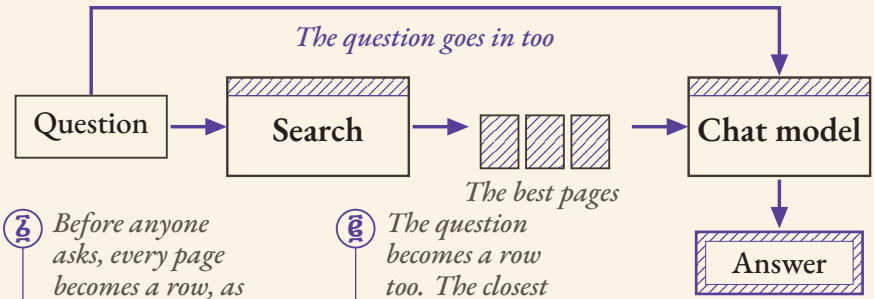
*Which pages?*

*The prompt*

What is our refund policy?

*In one study, the right pages added facts better than fine-tuning.*

# Search first, then answer from what was found



# The answer can point to its source

What is our refund policy?

You can return any item within a month for a full refund.



*Source: refund policy, returns*

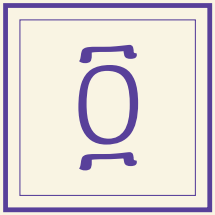
*The facts come from your pages, so you can check them.*

*The search can miss the right page. Check the source.*

*An invented answer*

# RAG runs beside the line





*Section 4*

# Human feedback

# People rank the answers

*You ask*

What is our refund policy?

*The model writes several answers*

- Check the website. — Second
- Refunds are possible, subject to the terms and conditions as published. — Third
- You can return any item within a month for a full refund. — Best

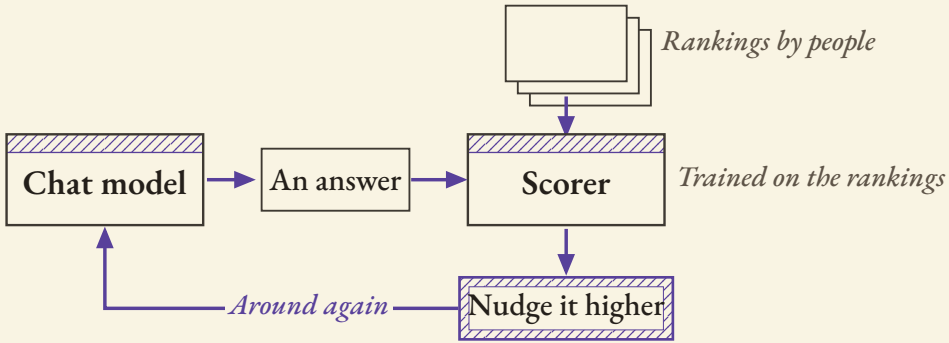


*A person ranks them*

*Invented answers*

*InstructGPT, 2022: 33 207 questions, each with 4 to 9 answers ranked.*

# A scorer learns the people's taste

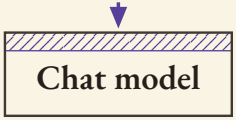



*Invented example*

# The model learns the taste of its raters



*The raters: about 40 contractors*

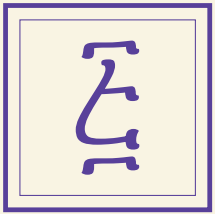



*In the study, the raters preferred the answers of a model 100 times smaller.*

*Approval can reward flattery: people sometimes prefer a convincing, agreeable answer to a correct one.*

# Three rungs up: answers people prefer



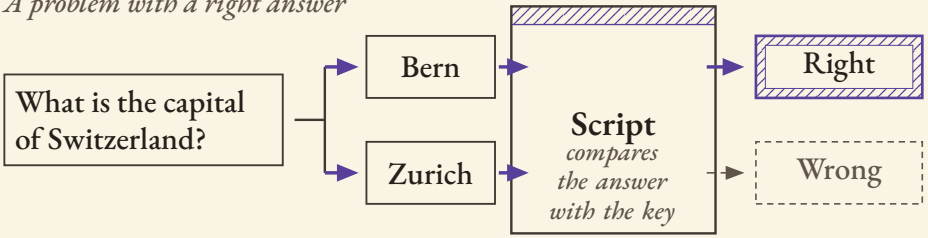


*Section 5*

# Verifiable rewards

# A script can check the answer

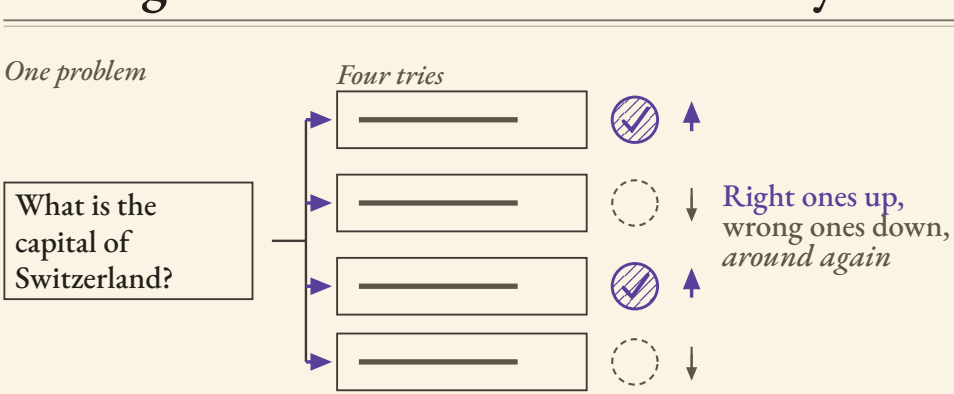
*A problem with a right answer*




*For code, the script runs the tests. A program that passes them is right.*

*Invented answers*

# The right answers become more likely




*Invented example*

# The model learns to think before it answers

A problem

Thinking

Answer

*Nobody wrote the steps. They appeared because they led to more right answers.*

*Right answers on a maths competition*

Before training 15.6%

After training 77.9%

# Four rungs up: it reasons before it answers



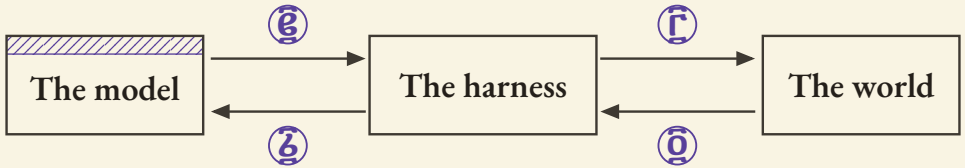
6

*Section 6*

# A harness

# The harness turns text into actions

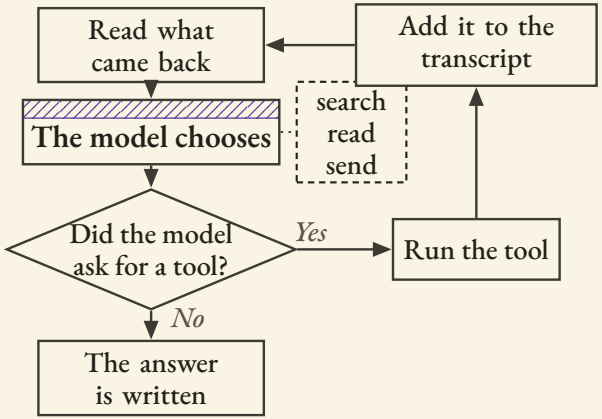
*The program around the model. It turns text into actions and feeds back what happened.*

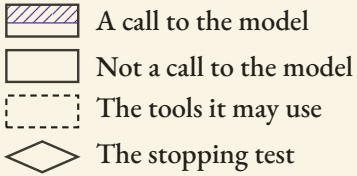



- Builds the prompt: the tools and the transcript
- Reads the reply and finds the tool call
- Runs that tool on the world
- Appends the result, then calls the model again or stops

*Anthropic, tool use, 2026*

# The loop: think, act, observe, again

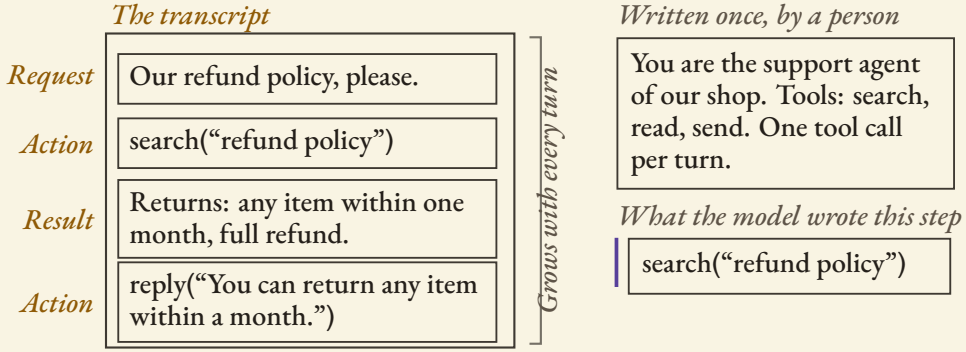


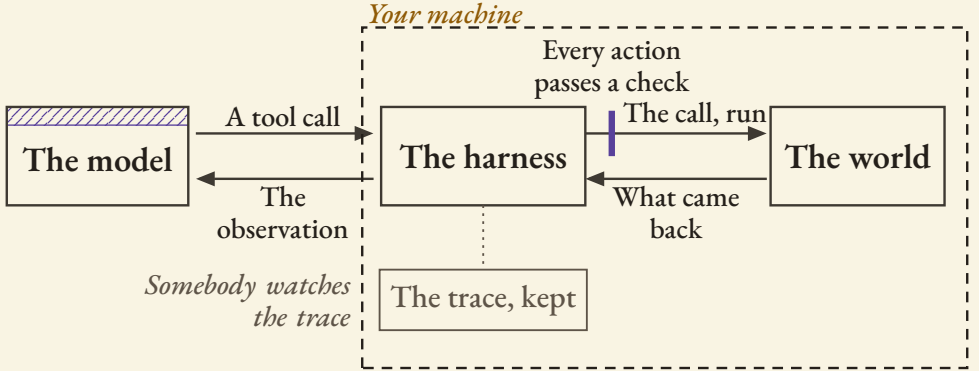

*The stopping test is in the loop's own condition. It is not a step counter.*

*Yao and others, ReAct, ICLR 2023*

# The transcript the model is called with



# An agent is a system




*Cherny, Claude Code, Anthropic, 2025*

# Five rungs up: it acts with tools



7

*Section 7*

# Planning strategies

# These tasks are games

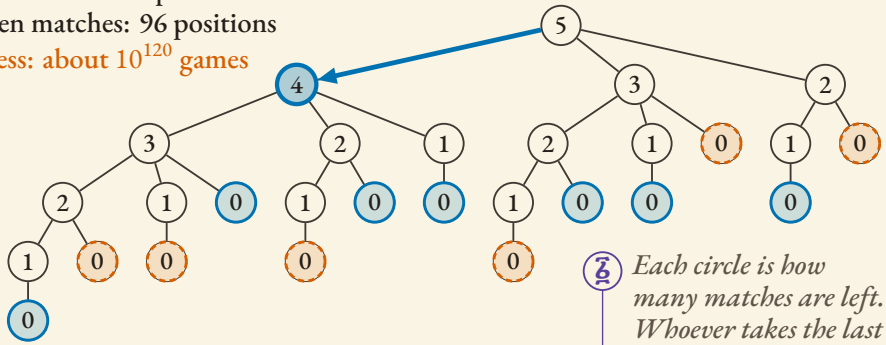
| The task | A position | A move | You have won when |
|---|---|---|---|
| **Chess** | The board right now | A legal move | Checkmate |
| **Fixing a bug** | The repository, and a failing test | An edit, or a command | The tests pass |
| **Booking a table** | The page you are on | A click, or something typed | The booking is confirmed |
| **Buying the right item** | The search results | A search, or a click on a product | It matches what was asked |



*In all four, one move is rarely right or wrong on its own. It is right if where it leads is good.*

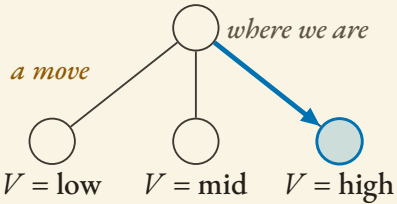
# The whole game tree, for once

Five matches: 28 positions
Seven matches: 96 positions
Chess: about $10^{120}$ games



# Two ways to score a position

*A critic: an opinion*




*A critic is fast, and it can be confidently wrong.*

*A rollout: play it out, at random*

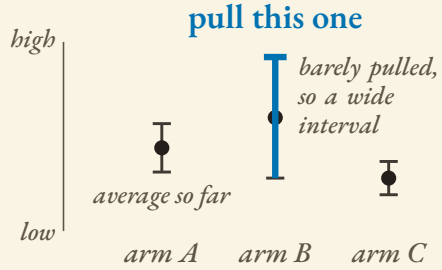



*The average of 100 random playouts*

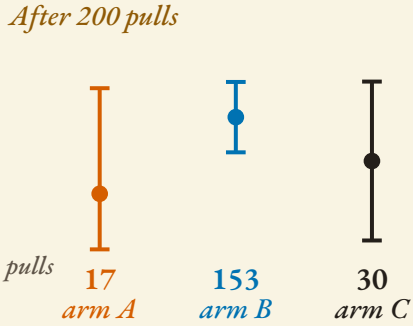
*A real measurement, one noisy sample at a time.*

*X to move*

# Pick the arm with the highest upper bound

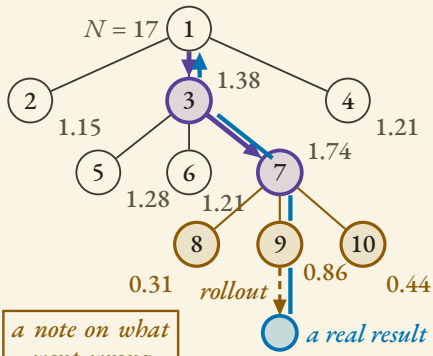



*A barely pulled arm has a wide interval, so it is tried even when its average is poor.*




*In the tree, this rule is UCT: how good it looked, plus how little we know.*

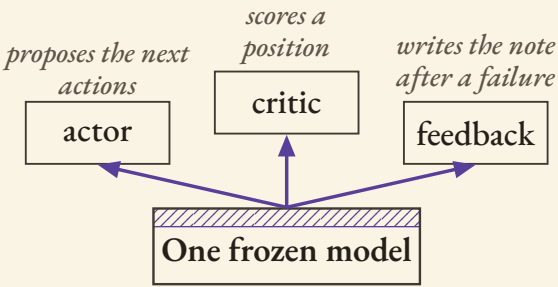
# The same loop, with a model in every box




| | | |
|---|---|---|
| Selection | *by UCT* | |
| Expansion | *the actor proposes* | |
| Evaluation | *the critic scores* | |
| Simulation | *the best one, to the end* | |
| Backpropagation | *the real result, up* | |
| 6 Reflection | *the model writes it down* | *into the next prompt* |

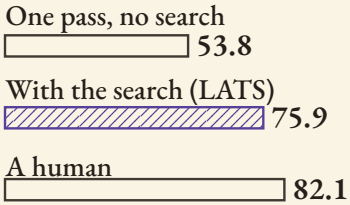
*Grey: UCT, what selection compares.*
*Ochre: V from the critic, and all that is new.*

# Three jobs, one set of weights




*The same weights every time.*
*Only the prompt changes.*

*A shopping site, 1.18 million real products*




*Zhou and others, LATS, ICML 2024*

# Six rungs up: it plans and reflects

